Muscle Proteomic Response to Blood Flow Restriction Training

Preliminary Analysis Report — BFR vs. HLRT

EDIT ME: Your Name

July 31, 2026

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Preliminary Analysis Report

Prepared for: **EDIT ME: PI Name**

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This report summarizes data preprocessing, quality control, normalization, differential abundance, and pathway enrichment analyses completed to date for the BFR vs. HLRT muscle proteomics study. Results are preliminary and subject to revision as the analysis pipeline continues.

Contents

Project Overview

EDIT ME: 1-2 paragraphs on study background/aims — e.g. blood flow restriction (BFR) vs. high-load resistance training (HLRT), what muscle adaptation question this proteomics dataset addresses, cohort size, timepoints (pre/post training), and tissue collected.

Study design at a glance:

Table 1: Final sample counts by group (post-QC).

Comparison	Contrast	Biological question
Baseline	BFR_PRE vs HLRT_PRE	Do the two arms differ before training begins?
Training_BFR	BFR_POST vs BFR_PRE	How does the muscle proteome change with BFR training?
Training_HLRT	HLRT_POST vs HLRT_PRE	How does the muscle proteome change with HLRT training?
Post_training	BFR_POST vs HLRT_POST	Do the two arms differ after training?
Interaction	(BFR_POST-BFR_PRE) vs (HLRT_POST-HLRT_PRE)	Does the training response itself differ by modality?

dataset	total_samples	n_rerun	n_BFR	n_HLRT
BFR	129	8	64	65

Pipeline Overview

The analysis pipeline proceeds through the following stages, each detailed in the sections below:

1. **Data preprocessing** — raw intensity/metadata loading, annotation cleanup, rerun consolidation
2. **Contaminant assessment** — tissue-specificity (HPA) and blood-protein flagging
3. **Filtering** — contaminant removal and per-group missingness filtering
4. **Outlier detection** — consensus-based sample QC
5. **Normalization** — cyclic loess normalization
6. **Differential abundance analysis** — limma, 5 contrasts
7. **Pathway / enrichment analysis** — GSEA + ORA across 7 databases, with redundancy filtering

1. Data Preprocessing

Raw protein-group intensities and sample metadata were loaded and restructured prior to any filtering:

- Protein annotation columns (UniProt ID, protein name, gene symbol, description) were parsed from the search-engine output, and multi-protein group matches were collapsed to their first (primary) match.
- cRAP (common Repository of Adventitious Proteins) contaminant tags were flagged directly from the raw protein group string, before any trimming could drop that information.
- Sample columns were cleaned to a consistent `S###` naming convention.
- **Rerun consolidation:** samples with a trailing "r"/"R" suffix (indicating a repeat MS injection) were identified and merged onto their base sample name, with a `rerun_level` flag retained in the metadata for traceability.
- Metadata and intensity columns were validated to be in 1:1 correspondence before proceeding (a hard `stopifnot` check — the pipeline halts rather than silently misaligning samples).

Table 3: Raw sample counts by group, prior to any QC or filtering.

group	n
BFR	65
HLRT	66

2. Contaminant Assessment

Two independent contamination checks were applied to every protein, in addition to a per-sample contamination index:

A. Tissue specificity (Human Protein Atlas). Each protein's UniProt ID was checked against the HPA skeletal muscle gene list; proteins absent from this reference were flagged as not muscle-specific.

B. Blood protein blacklist. A curated list of classic blood/plasma contaminant gene families was applied — albumin & lipid transport proteins, coagulation factors, complement system components, hemoglobins, acute-phase proteins, and platelet/neutrophil markers — supplemented dynamically with any immunoglobulin (IGH/IGK/IGL) or keratin (KRT) genes detected in this specific dataset.

C. Sample-level contamination index. For every sample, the summed intensity of blood markers (ALB, HBB, HBA1) vs. muscle markers (MB, CKM) was expressed as a percentage of total intensity, giving a blood:muscle (B:M) ratio per sample — a diagnostic for samples with unusually high blood contamination (e.g. from incomplete blood clearance during biopsy).

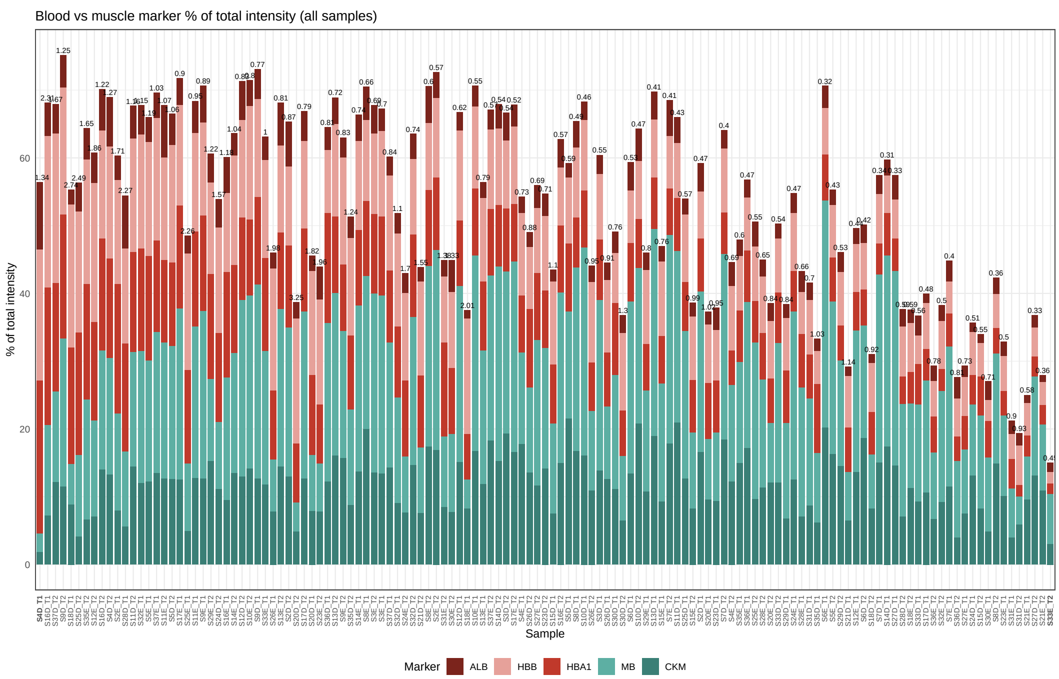

Figure 1: Blood vs. muscle marker proportion of total intensity, per sample.

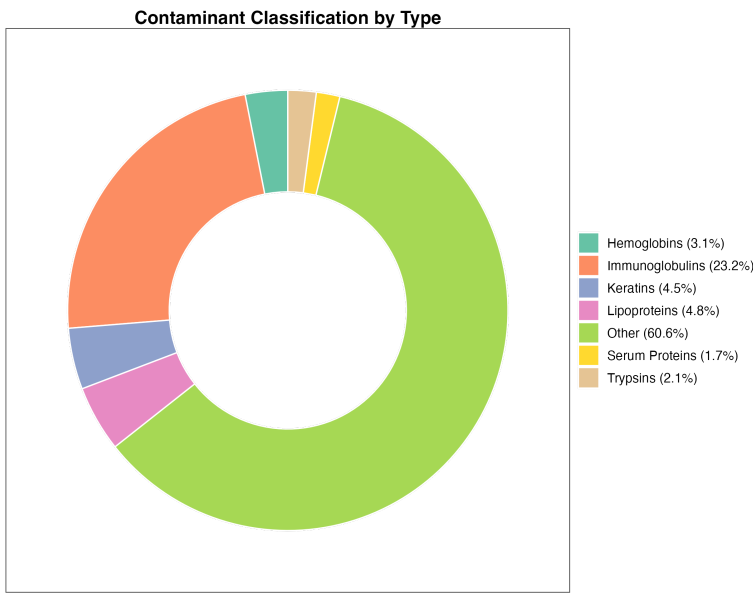

Figure 2: Classification of flagged contaminant proteins by category.

3. Filtering

Proteins and samples were filtered in the following order:

1. **Zero → missing.** Raw zero-intensity values (an artifact of the search engine, not a true measurement of zero abundance) were converted to `NA`.
2. **HPA muscle filter.** Proteins not annotated as muscle-expressed in the Human Protein Atlas were removed.
3. **Blood blacklist filter.** Proteins matching the blood contaminant gene list (Section 2B) were removed.
4. **Missingness filter.** Proteins were required to be observed in at least 50% of samples *within* at least one group (BFR or HLRT), rather than 50% of the full cohort — this retains proteins that are consistently detected in one arm even if largely absent in the other, which a whole-cohort threshold would have discarded.

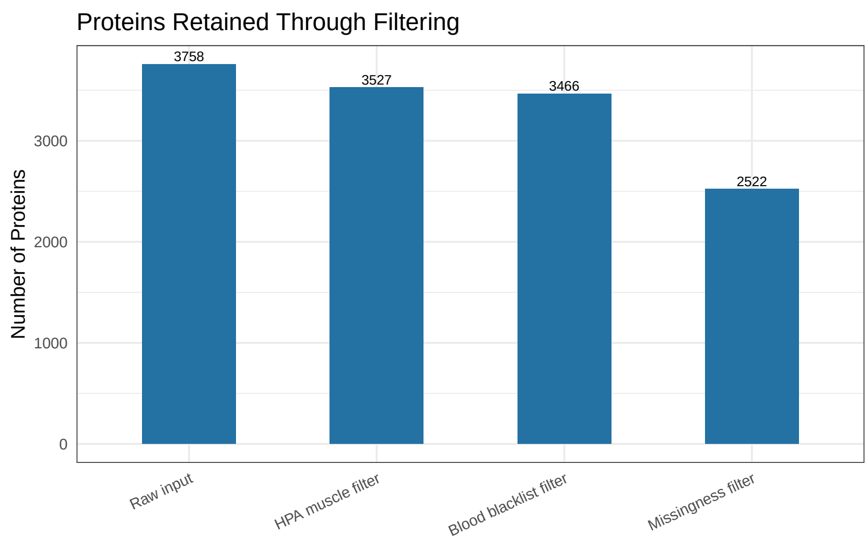

Figure 3: Number of proteins retained at each filtering step.

Table 4: Filtering effects: proteins before/after each step.

step	n_before	n_after	n_removed	pct_retained
Raw input	NA	3758	NA	100.0
HPA muscle filter	3758	3527	231	93.9
Blood blacklist filter	3527	3466	61	92.2
Missingness filter	3466	2522	944	67.1

4. Outlier Detection

Samples were screened for exclusion using a **consensus of three independent methods**, each flagging samples that are statistical outliers relative to the rest of the cohort:

- **Missingness outliers:** samples with % missing proteins beyond 1.5×IQR above the 75th percentile.
- **PCA outliers:** samples with a Mahalanobis distance (computed on the first 3 principal components of log2 intensities) exceeding the 99th-percentile χ^2 threshold.
- **Intensity outliers (MAD):** samples whose median log2 intensity deviates from the cohort median by more than 3× the median absolute deviation (MAD).

A sample was excluded only if flagged by **at least 2 of the 3** methods, to avoid removing samples based on a single noisy metric.

(showing first 20 of 131 rows)

Table 5: Full outlier diagnostic table (all samples, all metrics).

sample	pct_missing	miss_flag	mahal_dist	pca_flag	sample_median	mad_deviation	mad_flag	n_flags	consensus_outlier
S10D_T2	18945282	FALSE	1.6576999	FALSE	20.73892	0.1337513	FALSE	0	FALSE
S10D_T1	25860428	FALSE	1.9777415	FALSE	20.77555	0.0971280	FALSE	0	FALSE
S10E_T1	13481364	FALSE	2.4086022	FALSE	20.65404	0.2186402	FALSE	0	FALSE
S10E_T1	23877875	FALSE	1.4747740	FALSE	20.81909	0.0535810	FALSE	0	FALSE
S11D_T3	10134814	FALSE	1.8414668	FALSE	20.73066	0.1420189	FALSE	0	FALSE
S11D_T2	27755749	FALSE	1.5777259	FALSE	20.74714	0.1255308	FALSE	0	FALSE
S11E_T2	15773196	FALSE	1.7983743	FALSE	20.71398	0.1586977	FALSE	0	FALSE
S11E_T0	29119746	FALSE	3.2479640	FALSE	20.72234	0.1503342	FALSE	0	FALSE
S12D_T3	12117367	FALSE	1.7966916	FALSE	20.60964	0.2630343	FALSE	0	FALSE
S12D_T0	29119746	FALSE	1.9890576	FALSE	20.68003	0.1926450	FALSE	0	FALSE
S12E_T9	17938144	FALSE	1.4805216	FALSE	20.95305	0.0803734	FALSE	0	FALSE
S12E_T1	24.0761301	FALSE	0.5928723	FALSE	21.05590	0.1832217	FALSE	0	FALSE
S13D_T2	14187153	FALSE	2.6009233	FALSE	20.77957	0.0931094	FALSE	0	FALSE
S13D_T3	26478985	FALSE	1.0917658	FALSE	20.78754	0.0851386	FALSE	0	FALSE
S13E_T1	13.8382236	FALSE	1.0363993	FALSE	21.25350	0.3808216	FALSE	0	FALSE
S13E_T2	22601110	FALSE	2.6457615	FALSE	20.87268	0.0000000	FALSE	0	FALSE
S14D_T8	15646312	FALSE	0.7390484	FALSE	20.95303	0.0803559	FALSE	0	FALSE
S14D_T3	24496431	FALSE	1.2720878	FALSE	20.61863	0.2540455	FALSE	0	FALSE
S14E_T4	13219667	FALSE	1.0070278	FALSE	20.68003	0.1926450	FALSE	0	FALSE
S14E_T7	26130056	FALSE	1.0262073	FALSE	20.90976	0.0370893	FALSE	0	FALSE

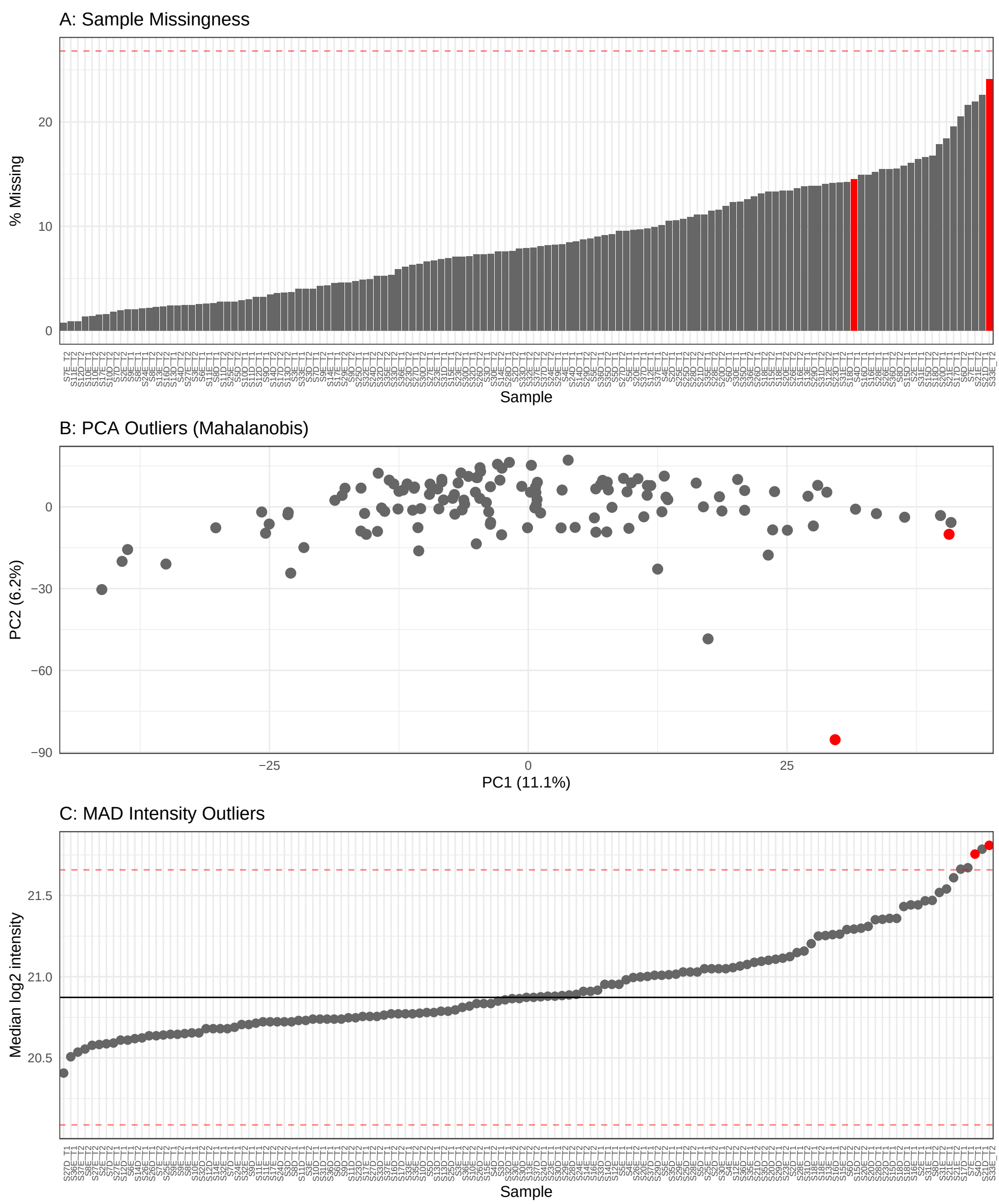

Figure 4: Outlier diagnostics: missingness, PCA (Mahalanobis), and MAD intensity.

5. Normalization

Following filtering and outlier removal, protein intensities were normalized using **cyclic loess normalization**, which iteratively aligns the intensity distribution of each sample against the others via local regression (loess) on pairwise MA-plots. This corrects for sample-to-sample technical variation (e.g. differences in total protein loaded, digestion/labeling efficiency) without assuming the majority of proteins are unchanged between the two study arms, which a simpler median/quantile normalization would.

Quality control before vs. after normalization:

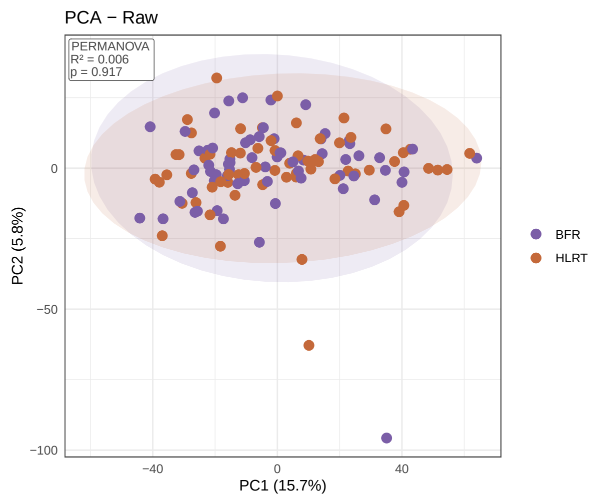

Figure 5: PCA of raw (pre-normalization) intensities, colored by group.

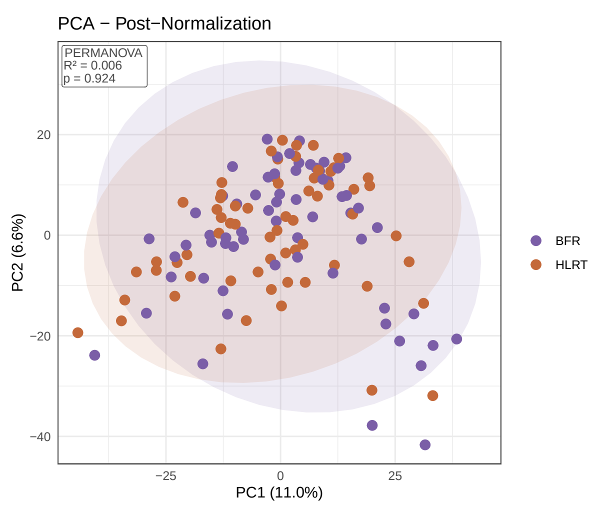

Figure 6: PCA of normalized intensities, colored by group.

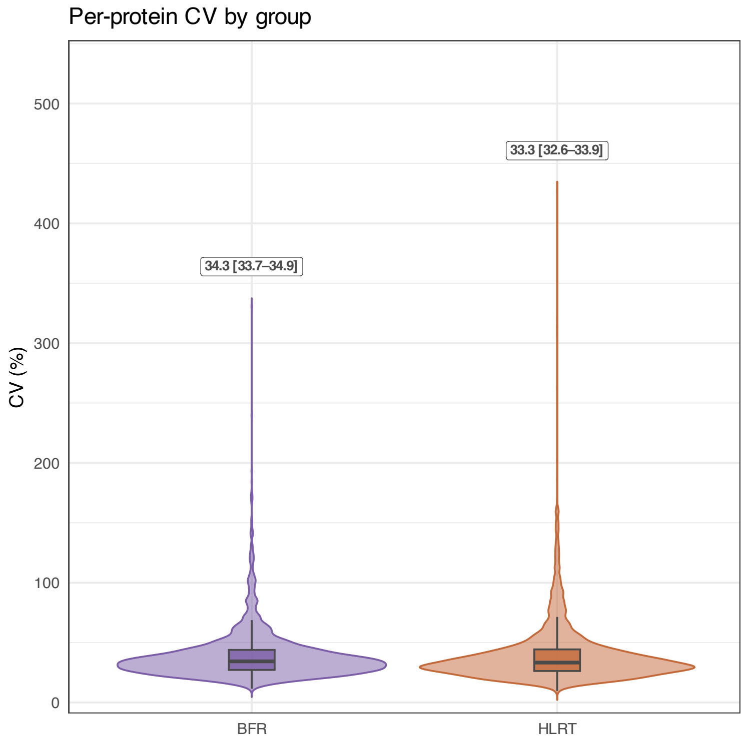

Figure 7: Per-protein coefficient of variation (CV) by group, pre-normalization.

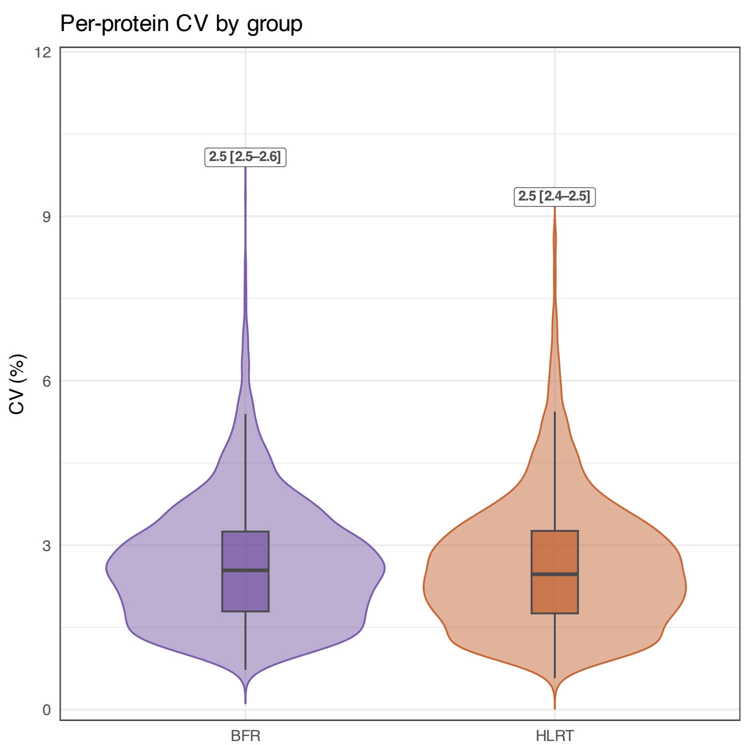

Figure 8: Per-protein coefficient of variation (CV) by group, post-normalization.

6. Differential Abundance Analysis

Differential abundance was assessed with **limma**, fit across 5 contrasts representing the study's 2$×$2 (arm × training) design plus the arm×time interaction (see the contrast table in Section "Project Overview"). For each contrast, three complementary significance metrics were retained: the raw p-value, the Benjamini-Hochberg FDR-adjusted p-value, and a composite **Pi-value** (EDIT ME: confirm/insert your exact Pi-value formula and rationale here — e.g. combining significance and effect-size magnitude) used as the primary threshold (Pi-value < 0.05) for calling a protein differentially abundant.

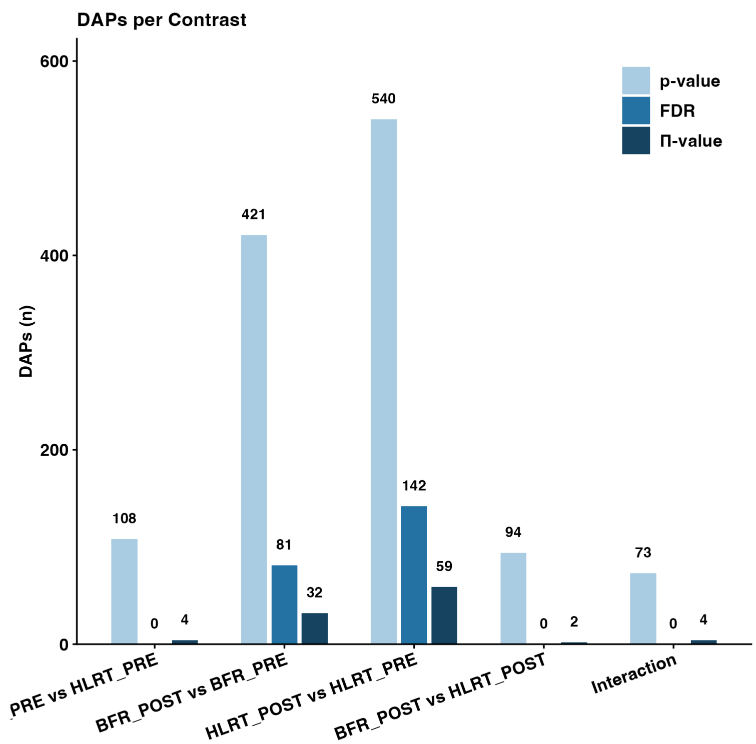

Figure 9: Number of differentially abundant proteins (DAPs) per contrast, by significance method.

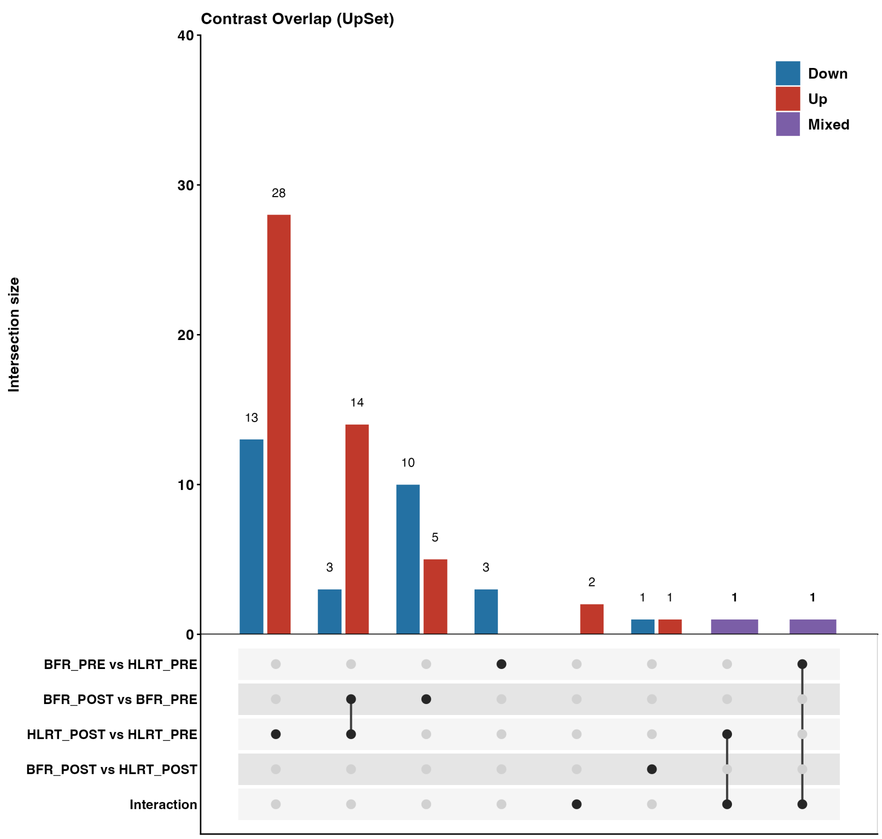

Figure 10: Overlap of differentially abundant proteins across contrasts (UpSet plot), colored by direction of change.

7. Pathway / Enrichment Analysis

For each of the 5 contrasts, two complementary enrichment approaches were run:

Gene Set Enrichment Analysis (GSEA) — rank-based, using the limma t-statistic as the ranking metric, tested against 7 pathway/gene-set databases: GO Biological Process, GO Cellular Component, GO Molecular Function (via `clusterProfiler::gseGO`), a GO Slim subset, KEGG, Reactome, and MSigDB Hallmark gene sets.

Over-Representation Analysis (ORA) — tested the sets of significantly up- and down-regulated proteins (Pi-value < 0.05) separately against GO BP/CC/MF, using all detected proteins as the statistical background.

Redundancy filtering. Because gene set databases contain many near-duplicate terms (e.g. highly overlapping GO terms), a pooled greedy redundancy filter was applied within each contrast $\times$ database: significant pathways were ranked by adjusted p-value, and a pathway was dropped if its gene-set membership overlapped an already-retained, more-significant pathway above a threshold (Jaccard index ≥ 0.5 **or** overlap coefficient ≥ 0.5). This keeps the most significant representative of each cluster of redundant terms rather than reporting near-duplicates as independent hits.

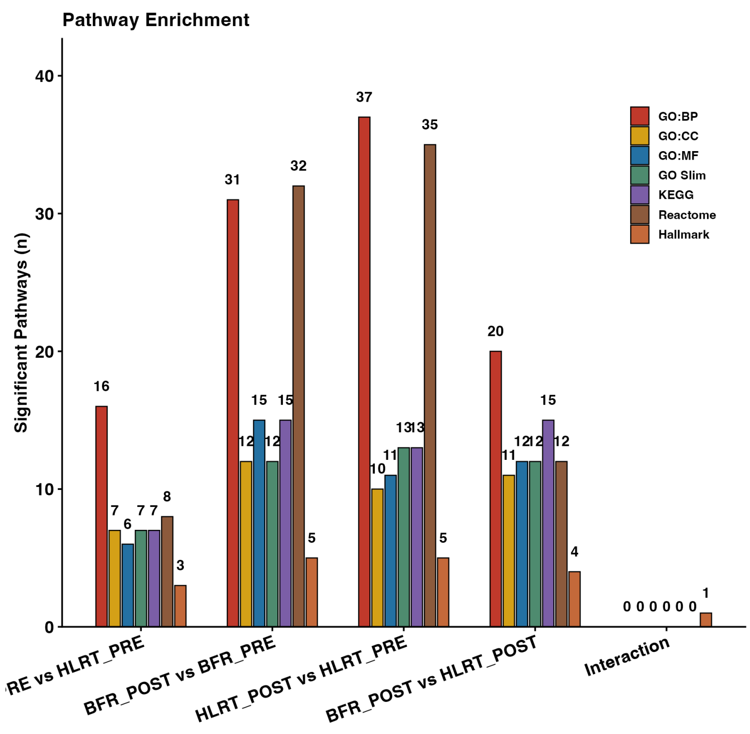

Figure 11: Number of significant (non-redundant) enriched pathways per contrast, by database.

8. Summary Figure

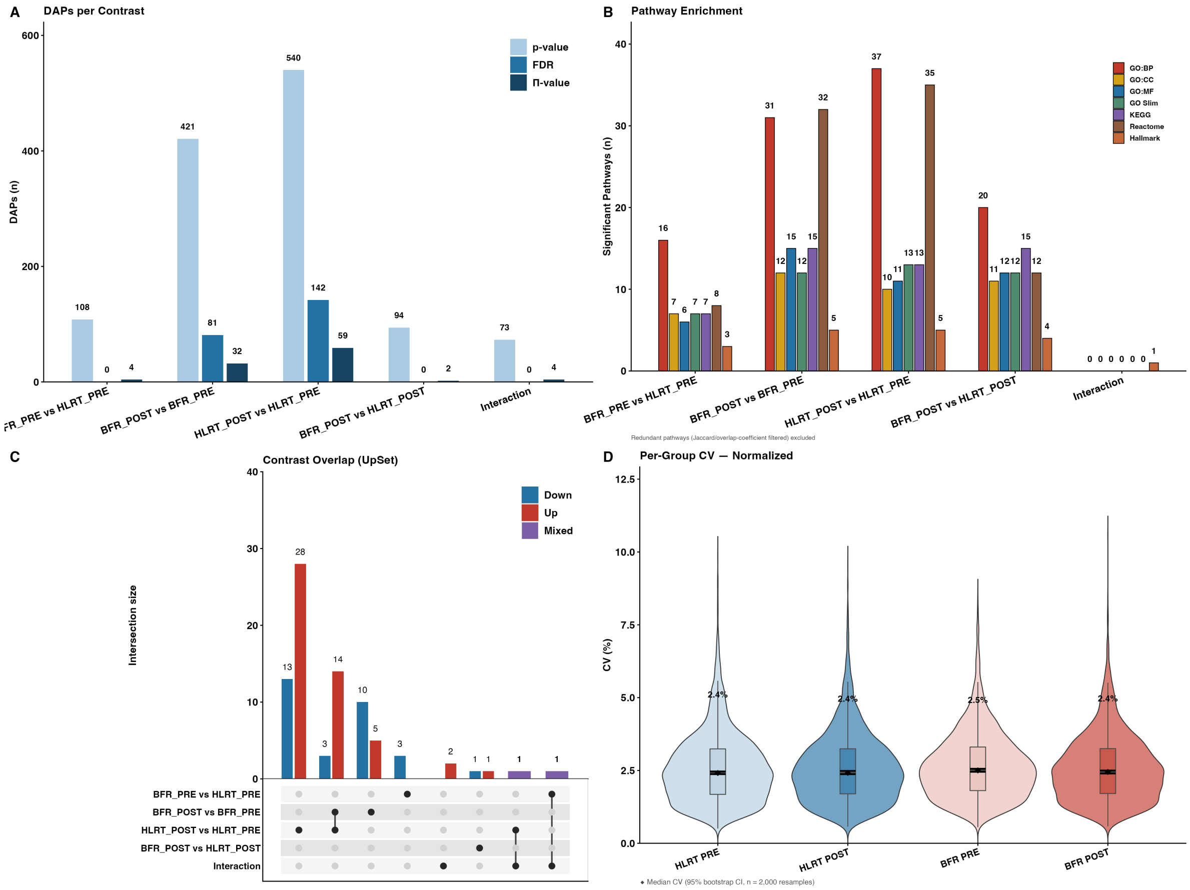

Figure 12: Combined overview: DAPs per contrast (A), pathway enrichment (B), contrast overlap (C), and per-group CV (D).

Next Steps

EDIT ME: bullet list of planned follow-up analyses — e.g. specific pathway deep-dives, targeted validation, additional contrasts, figures still to be generated for the manuscript, etc.